

YUVARAJ KESAVAN PREMKUMAR

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PROFESSIONAL SUMMARY

Data Science MSc candidate (University of Surrey, 2025–2026) with hands-on experience in machine learning, deep learning, NLP, and statistical modelling using Python. Skilled in building and evaluating supervised and unsupervised models (classification, regression, clustering, time series) on real-world datasets of 20,000–50,000+ records. Research experience in computer vision (EfficientNet-B0, Grad-CAM) with a peer-reviewed publication. Actively seeking entry-level / graduate Data Scientist or Machine Learning roles in the UK.

TECHNICAL SKILLS

Programming Languages: Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn, NLTK, OpenCV), SQL, Java, C

ML & Deep Learning: Supervised & Unsupervised Learning, Neural Networks, CNN, RNN, LSTM, GRU, Transfer Learning, EfficientNet, Grad-CAM, Model Evaluation (RMSE, MAE, F1, AUC-ROC), Cross-validation, Hyperparameter Tuning

NLP: Text Classification, Tokenisation, TF-IDF, N-gram Analysis, POS Tagging, LLM Integration (Mistral)

Data Analysis: EDA, Statistical Modelling, Feature Engineering, Data Cleaning, Data Wrangling, Hypothesis Testing

Databases: MySQL, MongoDB, BigQuery (familiar), Google Sheets

Visualisation & BI: Power BI, Matplotlib, Seaborn, Plotly

Cloud & Dev Tools: Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Flask, Google APIs

Soft Skills: Research & Analytical Thinking, Data Storytelling, Stakeholder Communication, Attention to Detail

EDUCATION

MSc Data Science — University of Surrey, Guildford, UK

Sep 2025 – Sep 2026 (Expected)

- Dissertation (in progress): AI-assisted detection of retinal biomarkers of Parkinson's disease using fundus and OCT images — planned approach: transfer learning with EfficientNet-B0, binary classification, and Grad-CAM explainability.
- Core modules: Machine Learning, Deep Learning, NLP, Statistical Modelling, Data Engineering, Intelligent Systems, Computer Vision.

BEng Computer Science and Engineering — PSG College of Technology, Coimbatore, India

Graduated Apr 2024

CGPA: 8.23 / 10 (82%)

WORK EXPERIENCE

Data Science Intern — KGiSL MicroCollege, Coimbatore, India

Mar 2025 – Sep 2025 (7 months)

- Built and evaluated machine learning models for text correction using Mistral LLM, integrating the model into a Flask-based full-stack application (HTML, CSS, JS).
- Applied LSTM and GRU deep learning models to time series regression tasks, comparing model performance using RMSE and MAE evaluation metrics.
- Conducted EDA and feature engineering on structured datasets using Python (pandas, NumPy) to identify patterns and prepare data for modelling.
- Built Power BI dashboards to visualise model outputs and statistical findings as actionable insights for stakeholders.

Software Development Intern — AppsComp Widgets Pvt Ltd, Coimbatore, India

Jan 2023 – Apr 2023 (4 months)

- Developed an e-commerce platform (Java, Python, HTML/CSS) handling 50,000+ product records with optimised MySQL database queries.
- Engineered an automated Python data analysis tool to process and report on 50,000+ entries, reducing manual reporting time significantly.
- Integrated data visualisation modules to convert 50,000+ data points into interactive insights for non-technical stakeholders.

Online Grocery Assistant (Part-time) — Sainsbury's, Godalming, UK

Nov 2025 – Present

- Maintained high accuracy (210 IPH) across 10+ daily picking runs in a fast-paced, data-tracked fulfilment environment, demonstrating reliability and attention to detail.

PROJECTS

Sign Language Recognition & Translation System (Final Year Project) — Python, CNN, scikit-learn, OpenCV *Sep 2023 – Mar 2024*

- Designed and trained a CNN-based image classification model on 20,000+ labelled gesture records, applying data augmentation and cross-validation to improve generalisation.
- Performed feature engineering and preprocessing (normalisation, resizing, augmentation) on image datasets using OpenCV and NumPy.
- Evaluated model performance using F1-score, confusion matrix, and accuracy metrics across 26 ASL signs; published findings in JUCCT, 2024.

Mental Health Disorder Trend Analysis & Dashboard — Python, pandas, scikit-learn, Power BI *Personal Project*

- Sourced, cleaned, and transformed public health datasets using Python (pandas), handling missing values, outliers, and normalisation before statistical analysis.
- Applied statistical trend modelling to identify patterns in global mental health disorder prevalence; visualised findings in a multi-page interactive Power BI dashboard.

AI-Powered Grammar Correction Chatbot — Python, Flask, Mistral LLM, NLP, HTML/CSS/JS *KGiSL Internship Project*

- Developed an end-to-end NLP application integrating the Mistral LLM to perform grammar correction on user-submitted text via a REST API (Flask backend).
- Designed prompt engineering strategies to optimise LLM output quality and consistency, and deployed a responsive chat interface for real-time interaction.

Location Tracking System — Python, Google Maps API *Jul 2022 – Dec 2022*

- Integrated Google Maps and Places APIs to build a secure employee location tracking system, implementing OAuth authentication workflows across 7 datasets.

Movie Streaming Platform — Java, MySQL, HTML/CSS *Jan 2023 – Aug 2023*

- Designed and implemented a relational database schema with intuitive UI/UX, supporting content browsing, user sessions, and search across a scalable web application.

PUBLICATION

- Sign Language Recognition and Translation System — Journal of Ubiquitous Computing and Communication Technologies (JUCCT), 2024.